

Solution for Automating Wire Harness Assembly

Hardware

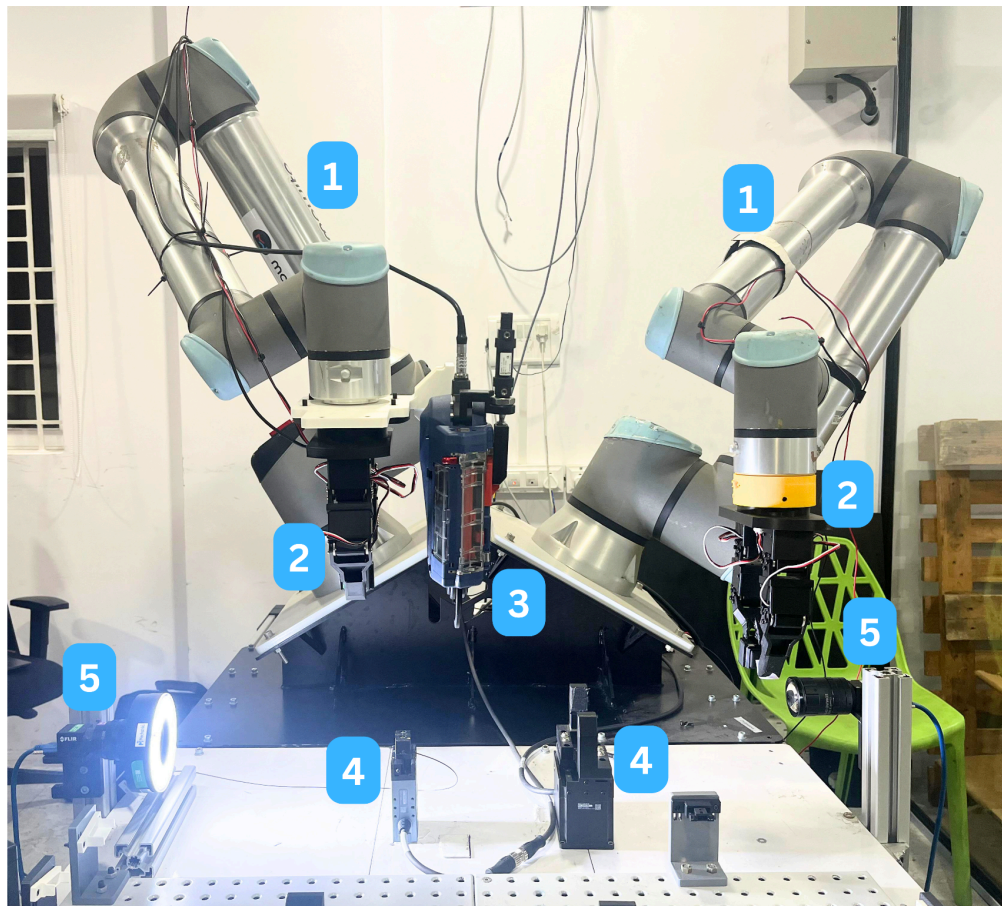


Image 1: Setup

The system (image 1) consists of the following hardware components:

1. Dual-cobot system
2. Neuralgripper - 2-finger (4 degrees of freedom) gripper with tactile sensor
3. Automatic clamping tool
4. Two-finger gripper
5. 2D camera

Video of how it works - [[VIDEO LINK](#)]

Innovations



Image 2: The NeuralGripper

1. NeuralGripper

NeuralGripper (image 2) is a custom-designed two-finger gripper developed primarily for precise in-hand manipulation of wires and cable crimps.

Each finger has **2 degrees of freedom**, achieved through two actuated revolute joints. Each joint is driven by a servo motor, allowing the gripper to manipulate the wire/crimp position with fine control and to use tactile feedback for contact awareness.

2. Model for in-hand manipulation for a precise pose of a given crimp

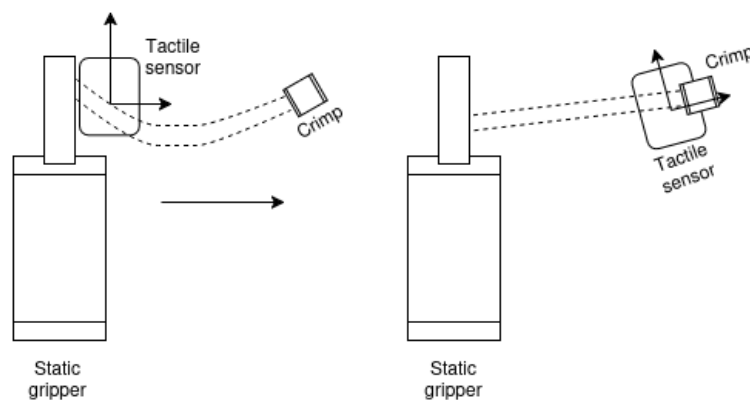


Figure 1: In-hand manipulation of the wire. On the left, in the initial state, the tactile sensor (at the tip of the robot's fingers) is holding a wire at its base. On the right, in the final state, the tactile sensor has moved to hold the wire crimp in the correct orientation (target pose) for insertion into the connector. [\[VIDEO LINK\]](#)

A Physical AI-based model was trained to achieve a precise target pose of a given wire crimp inside the NeuralGripper.

The model uses inputs from:

- **GelSight tactile sensors** [\[VIDEO LINK\]](#)
- **2D camera feedback**

Teleoperated demonstrations were collected to train the model. Approximately **400 minutes of training data** was collected across all four cable types.

The goal of the model is to predict the manipulation actions required to bring the wire crimp to a desired pose, enabling repeatable and precise in-hand alignment before the next assembly step.

Appendix

During teleoperation, the following constraints were maintained to ensure consistent and useful training data:

- A pulling force was maintained so that the wire remained under tension between the NeuralGripper and the fixed gripper.
- Grip force was controlled to maintain a clear tactile signature while keeping friction below a threshold that still allowed the wire to slide.
- The wire orientation was kept parallel to the centerline of the tactile sensor.
- At the end of each teleoperated demonstration, the wire end was required to be parallel to and centered on the tactile sensor.

A **PD controller** was implemented during teleoperation to regulate the grip force and keep it within the desired range. This helped maintain stable tactile contact while still allowing controlled sliding of the wire through the gripper.